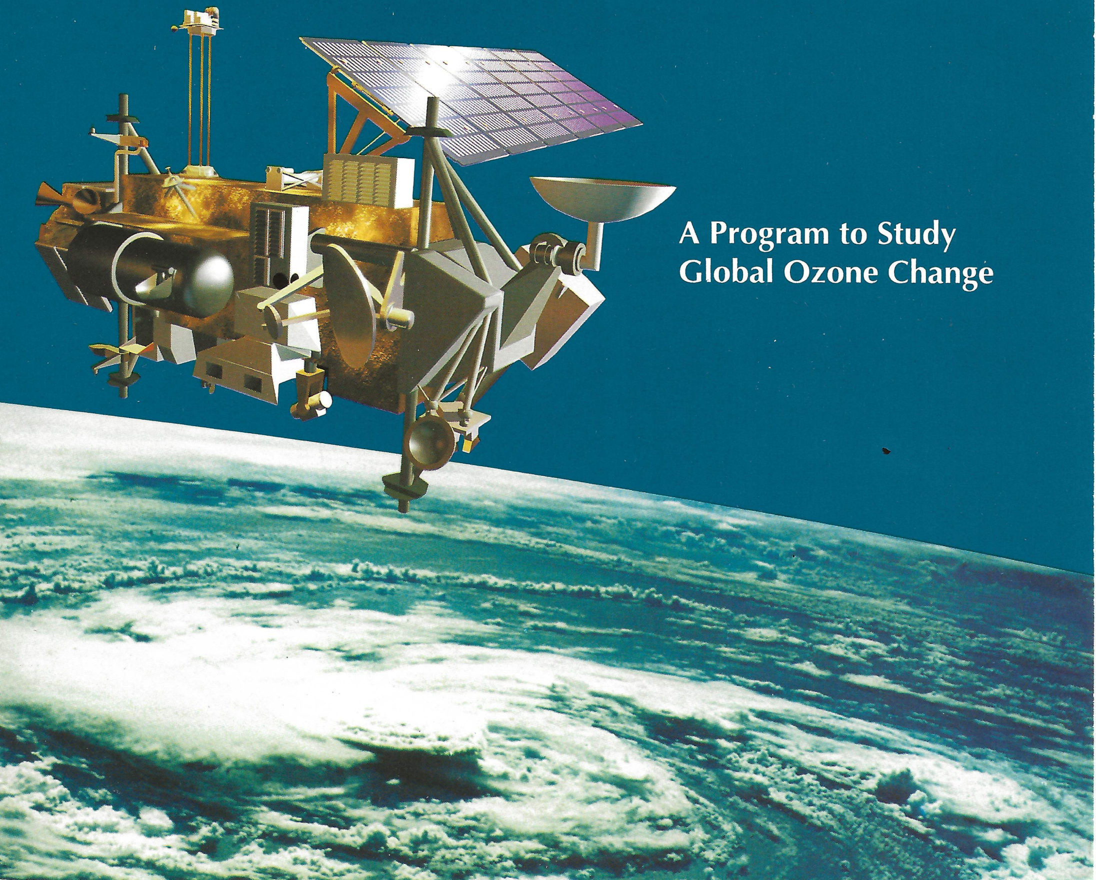
UARS

Upper Atmosphere Research Satellite

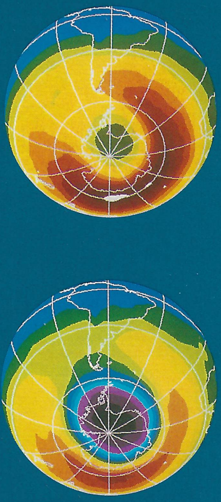


A Program to Study
Global Ozone Change

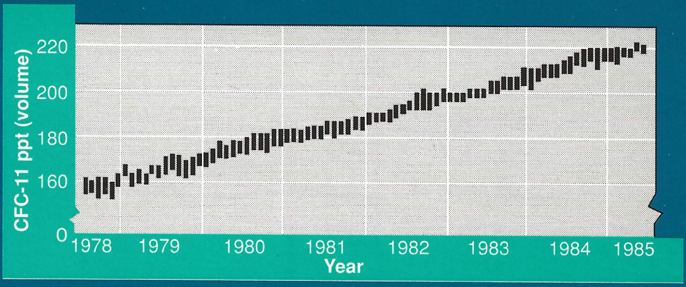
THE EARTH'S UPPER ATMOSPHERE

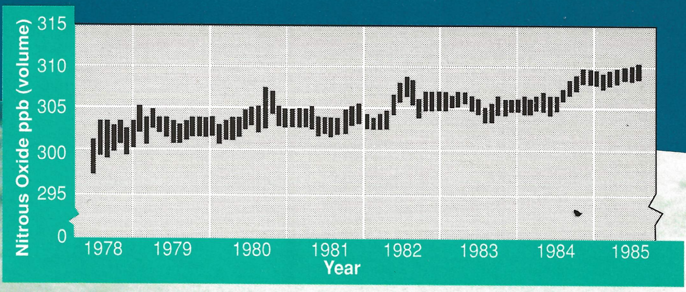
It's thin, fragile, almost invisible. Yet it plays a fundamental role in global climate and provides our shield against harmful ultraviolet radiation from the sun.

This shield—the stratospheric ozone layer—may be failing. The release of industrial chlorofluorocarbons, or CFCs, is almost certainly responsible for the annual depletion of stratospheric ozone (the "ozone hole") observed over Antarctica since 1979. And ozone thinning has now been found over the Arctic as well.



Growth of Antarctic ozone hole: average October ozone concentrations for 1979 (top) and 1989 (bottom) mapped from space by NASA's Total Ozone Mapping Spectrometer. Violet, blue: low ozone levels. Green, reds: higher levels.





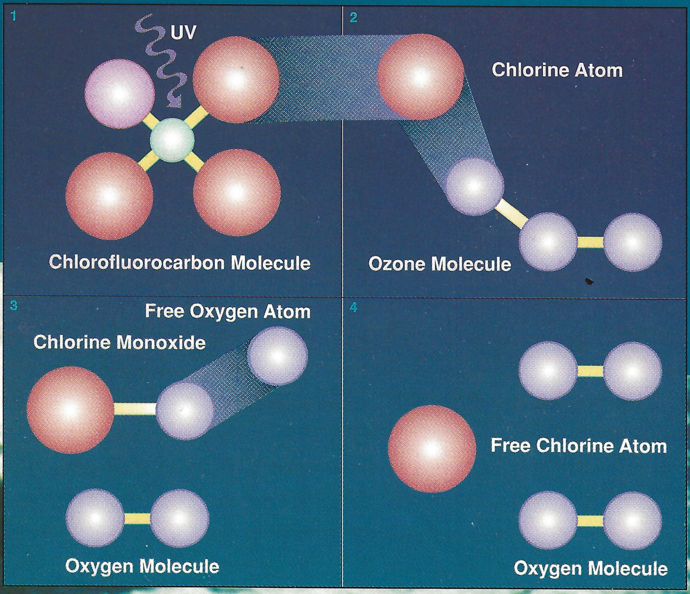
Atmospheric concentrations of many chemically important trace species continue to rise, including CFC-11 (top) and nitrous oxide (bottom).

A CRUCIAL FRONTIER FOR SPACE RESEARCH

Human societies are now capable of changing the properties of the Earth's atmosphere on a global scale. One serious concern is high altitude ozone depletion—caused in part by human activities such as the release of nitrogen, hydrogen and chlorine compounds into the atmosphere. The stratospheric ozone layer, 15 to 30 miles above the surface, screens out high-energy solar ultraviolet radiation that can cause skin cancer in human beings as well as damage to other animal and plant life.

Recently documented ozone depletion over the Earth's polar regions may represent just the beginning of global ozone loss. Ozone levels and other upper-atmosphere properties are determined by interactions among three key processes—chemistry, dynamics, and energy input—that are inherently global in scope. Only remote sensing from space can provide the temporal and spatial coverage needed to study such highly variable processes worldwide.

That's why NASA's Upper Atmosphere Research Satellite (UARS) is so important. The UARS mission will provide a dramatic step forward in our understanding of the upper atmosphere—enabling governments around the world to assess the role of human activities in stratospheric ozone depletion and to develop informed policies in response.



Chlorine molecules, freed from CFCs by solar ultraviolet radiation, destroy stratospheric ozone through catalytic chemical reactions.

THE UARS PROGRAM

The UARS mission will carry nine complementary experiments, each providing measurements critical to an understanding of upper-atmosphere processes.

Chemistry. Four instruments (CLAES, ISAMS, MLS, HALOE) will measure temperature profiles and concentrations of ozone, methane, water vapor, and other key trace species, including the most important CFCs.

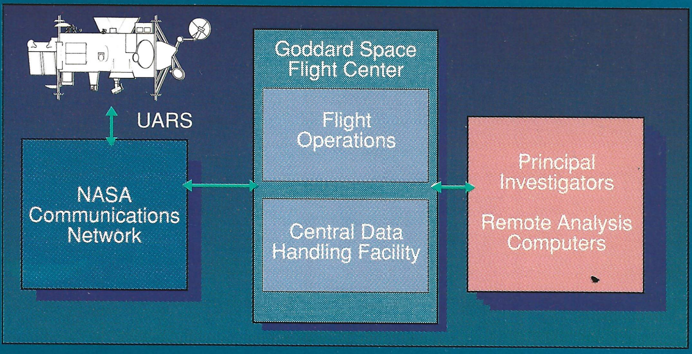
Dynamics. Two instruments (HRDI, WINDII) will map upper-atmosphere wind fields, which shape the global distribution of chemical species, through Doppler-shift measurements using high resolution interferometry.

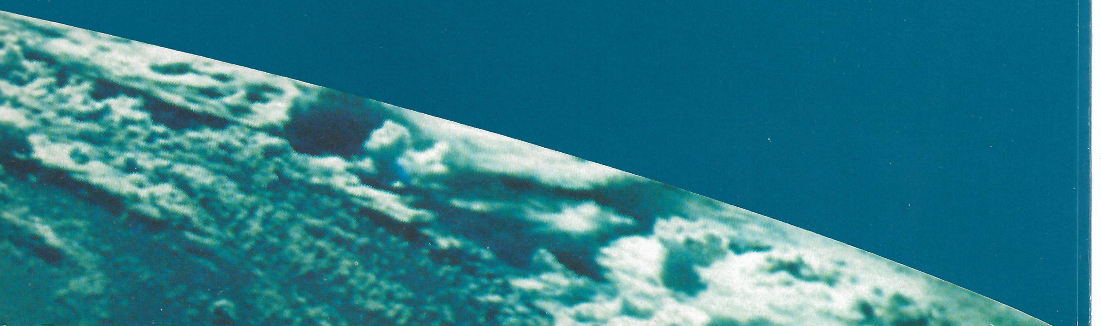
Energy Input. Three instruments (SUSIM, SOLSTICE, PEM) will determine energy inputs from the Sun and the Earth's magnetosphere, which strongly influence both chemistry and dynamics.

An additional instrument will extend long-term measurements of the solar constant.

The UARS program will also support ten theoretical groups with specific responsibilities for data analysis and interpretation. And an advanced data system will permit data analysis and selection of observing targets to be carried out remotely from computer terminals at investigators' home institutions.

UARS pioneers the integration of flight operations, communications, data processing, and scientific data analysis into a single, unified experiment.





NASA'S UPPER ATMOSPHERE RESEARCH SATELLITE (UARS)

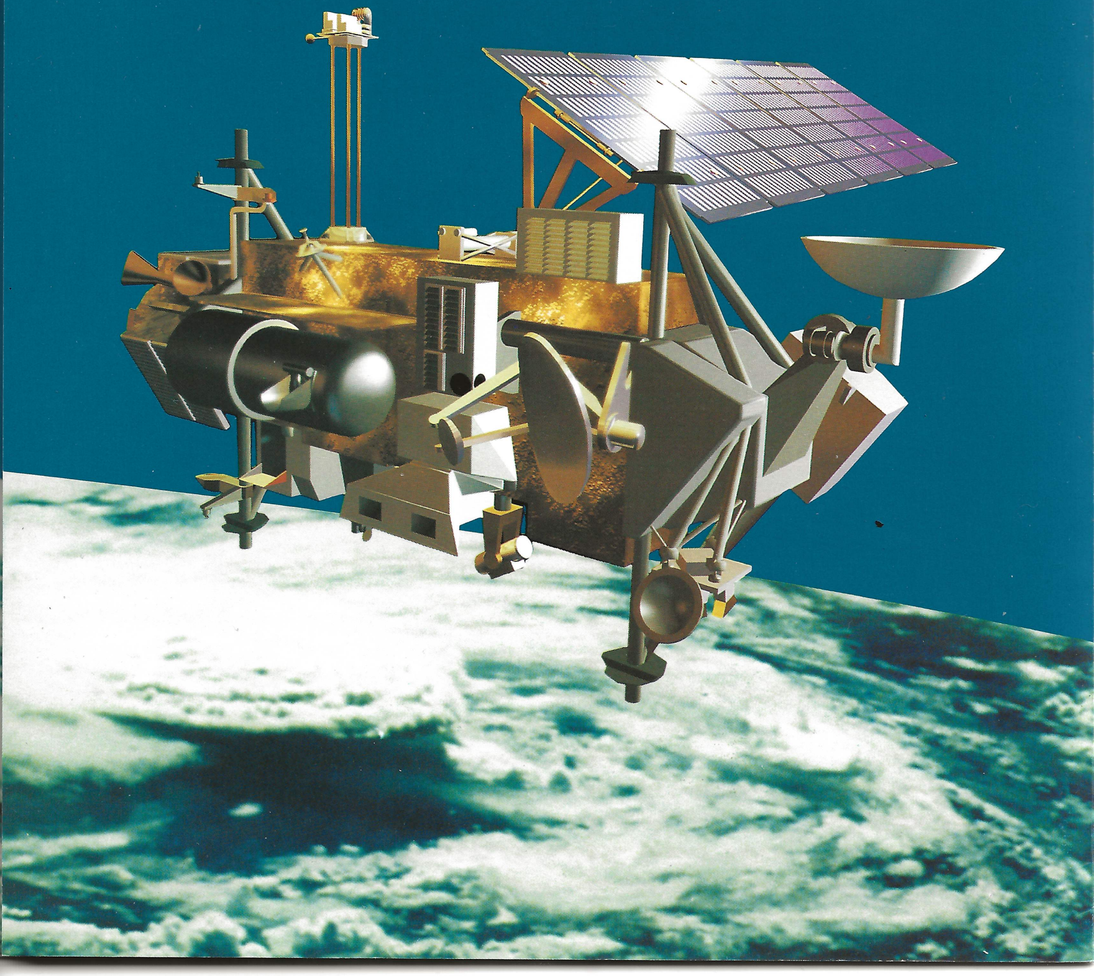
UARS is the spearhead of a long-term, international program of space research into global atmospheric change. Under United States leadership, the UARS program will:

• Carry out the first systematic, detailed satellite study of the Earth's stratosphere, mesosphere, and lower thermosphere;

• Establish the comprehensive data base needed for an understanding of stratospheric ozone depletion; and

• Bring together scientists and governments around the world to assess the role of human activities in atmospheric change.

Scheduled for launch in 1991, the UARS mission will also lay the foundations for a broader study of upper-atmosphere influence on climate and climatic variations.



UARS INSTRUMENTS AND PRINCIPAL INVESTIGATORS

CLAES: Cryogenic Limb Array Etalon Spectrometer (infrared emission spectrometer). A.E. Roche, Lockheed Palo Alto Research Laboratory. Collaborative investigator: J. Gille, National Center for Atmospheric Research.

ISAMS: Improved Stratospheric and Mesospheric Sounder (infrared emission radiometer). F.W. Taylor, Oxford University. Collaborative investigator: J.M. Russell, NASA Langley Research Center.

MLS: Microwave Limb Sounder (microwave emission radiometer). J.W. Waters, Jet Propulsion Laboratory.

HALOE: Halogen Occultation Experiment (infrared absorption radiometer). J.M. Russell, NASA Langley Research Center.

HRDI: High Resolution Doppler Imager (Fabry-Perot visible and near-infrared emission and absorption interferometer). P.B. Hays, University of Michigan.

WINDII: Wind Imaging Interferometer (Michelson visible and near-infrared emission interferometer). G.G. Shepherd, York University.

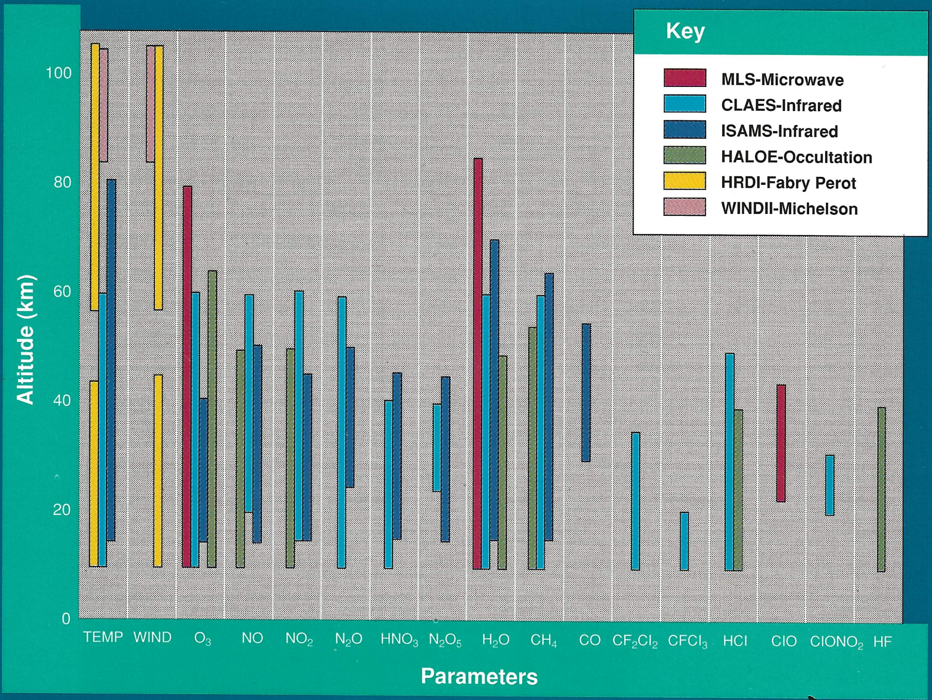
SUSIM: Solar Ultraviolet Spectral Irradiance Monitor (solar ultraviolet spectrometer). G.E. Brueckner, Naval Research Laboratory.

SOLSTICE: Solar Stellar Irradiance Comparison Experiment (solar ultraviolet spectrometer). G.J. Rottman, University of Colorado.

PEM: Particle Environment Monitor (electron, proton, and imaging x-ray spectrometer). J.D. Winningham, Southwest Research Institute.

UARS THEORETICAL INVESTIGATIONS AND PRINCIPAL INVESTIGATORS

Impact of ozone change on dynamics	D.M. Cunnold	Georgia Institute of Technology
Dynamics	M. Geller	State University of New York
Transports, budgets, and energetics	W.L. Grose	NASA Langley Research Center
Wave dynamics and transport	J. R. Holton	University of Washington
Response to solar variations	J. London	University of Colorado
Meteorological interpretation	A.J. Miller	National Oceanic and Atmospheric Administration
Analytic-empirical modeling	C.A. Reber	NASA Goddard Space Flight Center
Three-dimensional stratospheric model	P. White	United Kingdom Meteorological Office
Chemical, dynamic, and radiative processes	D. Wuebbles	Lawrence Livermore National Laboratory
Radiative-dynamic balance	R.W. Zurek	Jet Propulsion Laboratory



UARS chemistry and dynamics sensors will make measurements of temperature, pressure, wind velocity, and gas species concentrations in the altitude ranges shown above.

For More Information Contact:
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NASA

Credits: UARS, Angel Studios; Hurricane, NASA; Ozone, M. Schoeberl, GSFC/NASA.